

A 79-VERTEX COUNTEREXAMPLE TO CONJECTURE 4.1 OF HURLBERT AND KAMAT

ABSTRACT. We exhibit a tree on 79 vertices with no vertex of degree two such that, for $r = 18$, a degree-three vertex is contained in more independent r -sets than any leaf. Thus the tree is not 18-HK, disproving Conjecture 4.1 of Hurlbert and Kamat. The comparison is certified by exact independence-polynomial identities.

1. INTRODUCTION

For a graph G and $1 \leq r \leq \alpha(G)$, let $\mathcal{I}^r(G)$ be the family of independent r -subsets of $V(G)$, and let

$$\mathcal{I}_v^r(G) = \{A \in \mathcal{I}^r(G) : v \in A\}.$$

A vertex v is a *maximum r -center* if $|\mathcal{I}_v^r(G)|$ is maximum over $V(G)$. Following Estrugo and Pastine [2], a tree is r -HK if some leaf is a maximum r -center, and is HK if it is r -HK for every admissible r .

Counterexamples to the unrestricted assertion that every tree is HK are known [1, 3]. Hurlbert and Kamat subsequently conjectured that every tree with no vertex of degree two is HK [5, Conjecture 4.1]; see also [4, Section 4]. We give a counterexample.

Theorem 1. *There is a tree T on 79 vertices, with no vertex of degree two, and a vertex c of degree three such that*

$$|\mathcal{I}_c^{18}(T)| = 1,526,033,336,050,350 > \max_{u \text{ a leaf of } T} |\mathcal{I}_u^{18}(T)| = 1,524,522,945,307,798.$$

Consequently, T is not 18-HK, and Conjecture 4.1 of Hurlbert and Kamat is false.

2. CONSTRUCTION AND VERIFICATION

Proof of Theorem 1. Let $t_1 = 9$ and $t_2 = t_3 = 8$. Define T by

$$\begin{aligned} V(T) = & \{c\} \cup \{h_i : 1 \leq i \leq 3\} \cup \{s_{ij} : 1 \leq i \leq 3, 1 \leq j \leq t_i\} \\ & \cup \{\ell_{ijk} : 1 \leq i \leq 3, 1 \leq j \leq t_i, k \in \{1, 2\}\}, \end{aligned}$$

and with edges

$$ch_i, \quad h_i s_{ij}, \quad s_{ij} \ell_{ijk}$$

2020 *Mathematics Subject Classification.* 05C69, 05D05.

Key words and phrases. independent set, maximum star center, HK tree, tree, Erdős–Ko–Rado.

for all admissible indices. Every vertex other than c has a unique neighbor closer to c , so T is a tree. Moreover,

$$|V(T)| = 1 + 3 + 3(t_1 + t_2 + t_3) = 79.$$

Also,

$$\deg(c) = \deg(s_{ij}) = 3, \quad \deg(h_i) = t_i + 1 \in \{9, 10\}, \quad \deg(\ell_{ijk}) = 1,$$

so T has no vertex of degree two. The three vertices h_i together with all 50 leaves form an independent set, so $18 \leq \alpha(T)$.

For a graph F , write

$$P_F(z) = \sum_{k \geq 0} i_k(F) z^k,$$

where $i_k(F)$ is the number of independent k -sets of F . For every graph G and vertex $v \in V(G)$,

$$(1) \quad |\mathcal{I}_v^r(G)| = [z^{r-1}] P_{G-N_G[v]}(z),$$

where $N_G[v]$ is the closed neighborhood of v .

A support vertex s_{ij} together with its two leaf neighbors induces a three-vertex path, whose independence polynomial is

$$S(z) = 1 + 3z + z^2.$$

For $t \geq 0$, set

$$H_t(z) = S(z)^t + z(1+z)^{2t}.$$

This is the independence polynomial of a branch with one hub, t support vertices, and two leaves adjacent to each support vertex: the first term omits the hub, and the second includes it.

Deleting $N_T[c]$ leaves 25 disjoint three-vertex paths, and hence

$$(2) \quad P_{T-N_T[c]}(z) = S(z)^{25}.$$

The leaves form two orbits. Let u_8 be a leaf in a branch with eight support vertices, and let u_9 be a leaf in the branch with nine support vertices. After deleting a leaf and its support vertex, its sibling remains isolated. Splitting the remaining graph according to whether c is chosen gives

$$(3) \quad P_{T-N_T[u_8]}(z) = (1+z)(zS(z)^{24} + H_9(z)H_8(z)H_7(z)),$$

$$(4) \quad P_{T-N_T[u_9]}(z) = (1+z)(zS(z)^{24} + H_8(z)^3).$$

For nonnegative integers q, m , with the convention $\binom{n}{k} = 0$ for $k < 0$ or $k > n$, exact coefficients are obtained from

$$(5) \quad [z^m]S(z)^q = \sum_{j=\max\{0, m-q\}}^{\lfloor m/2 \rfloor} \binom{q}{j} \binom{q-j}{m-2j} 3^{m-2j}$$

and

$$[z^m]H_t(z) = [z^m]S(z)^t + \binom{2t}{m-1}.$$

Applying (5) and ordinary coefficient convolution to (2)–(4) yields

$$[z^{17}]P_{T-N_T[c]}(z) = 1,526,033,336,050,350,$$

$$[z^{17}]P_{T-N_T[u_8]}(z) = 1,524,522,945,307,798,$$

$$[z^{17}]P_{T-N_T[u_9]}(z) = 1,521,164,183,552,880.$$

In particular,

$$|\mathcal{I}_c^{18}(T)| - |\mathcal{I}_{u_8}^{18}(T)| = 1,510,390,742,552 > 0.$$

By (1), the star centered at c is larger than every leaf-centered 18-star. Hence no leaf is a maximum 18-center, and T is not 18-HK. $\square$

Remark. No minimality claim is made for the order of the construction.

REFERENCES

- [1] P. Borg, *Stars on trees*, Discrete Math. **340** (2017), no. 5, 1046–1049. doi:10.1016/j.disc.2016.11.002.
- [2] E. J. J. Estrugo and A. Pastine, *On stars in caterpillars and lobsters*, Discrete Appl. Math. **298** (2021), 50–55. doi:10.1016/j.dam.2021.03.013.
- [3] C. Feghali, M. Johnson, and D. Thomas, *Erdős–Ko–Rado theorems for a family of trees*, Discrete Appl. Math. **236** (2018), 464–471. doi:10.1016/j.dam.2017.10.009.
- [4] G. Hurlbert, *A survey of the Holroyd–Talbot conjecture*, in G. O. H. Katona, B. Patkós, and C. Tompkins (eds.), *Sum(m)it280: Surveys in Extremal Combinatorics and Combinatorial Geometry*, Bolyai Society Mathematical Studies, vol. 32, Springer, Cham, 2026, pp. 183–201. doi:10.1007/978-3-032-18810-6_10.
- [5] G. Hurlbert and V. Kamat, *On intersecting families of independent sets in trees*, Discrete Appl. Math. **321** (2022), 4–9. doi:10.1016/j.dam.2022.06.028.